



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	MD Razia
Student Name	Ritwik
Roll Number	2520080051
Branch	AI&DS
Course Outcome	Co-4

WaterGrid – Smart Water Distribution & Monitoring System

Work Done Summary:

Building on the WaterGrid Smart Water Distribution & Monitoring System, this case study extends the network analysis from Minimum Spanning Tree construction to shortest-path computation. Three classical shortest-path algorithms, Dijkstra's Algorithm, the Bellman-Ford Algorithm, and the Floyd-Warshall Algorithm, were implemented and compared on the same WaterGrid pipeline network to determine the most efficient route for water flow and monitoring signals between distribution hubs. Water distribution hubs were modelled as graph vertices, and pipelines were modelled as directed, weighted edges representing pumping and flow cost, including one negative-weight edge representing a gravity-fed pipeline segment that reduces effective pumping cost. Each algorithm was applied to compute shortest delivery paths, and the results were compared for correctness, efficiency, and applicability across different graph types (weighted/unweighted, directed/undirected, and graphs containing negative edge weights). The comparison confirmed that Bellman-Ford and Floyd-Warshall both correctly computed a minimum pipeline cost of 7 units from Hub A to Hub D by using the gravity-fed segment, while Dijkstra's Algorithm, which cannot safely process negative weights, had to be restricted to the non-negative subgraph to avoid incorrect results.

Implementation Details:

1. Represented WaterGrid water distribution hubs as graph vertices, consistent with the Co-3 MST network model.
2. Represented pipeline connections as directed, weighted edges with pumping and flow cost, including a gravity-fed segment with negative cost.
3. Applied Dijkstra's Algorithm using a min-priority queue to compute single-source shortest paths on the non-negative weighted subgraph.
4. Applied the Bellman-Ford Algorithm by relaxing all edges $(|V| - 1)$ times to correctly handle the negative-weight gravity-fed segment and to detect any negative-weight cycles.
5. Applied the Floyd-Warshall Algorithm using dynamic programming to compute all-pairs shortest paths across every hub in the WaterGrid network.

6. Compared the three algorithms on correctness, time complexity, and suitability for weighted/unweighted, directed/undirected, and negative-weight pipeline graphs.

7. Identified the minimum-cost flow path and validated that Bellman-Ford and Floyd-Warshall produced identical, correct shortest-path costs across the WaterGrid network.

Results and Screenshots:

Input Pipeline Graph, Dijkstra's Algorithm and Bellman-Ford Algorithm Output:

```
WATERGRID - SHORTEST PATH ANALYSIS
DIJKSTRA / BELLMAN-FORD / FLOYD-WARSHALL
-----

Input WaterGrid Pipeline Graph (Directed, Weighted):
A --6--> B
A --4--> C
B --5--> D
C --2--> B    (gravity-fed segment)
C --3--> D
D --2--> E
B --7--> E

1. Dijkstra (positive subgraph only):
Note: pipeline C->B excluded, Dijkstra needs w >= 0
A->A=0  A->B=6  A->C=4  A->D=7  A->E=9

2. Bellman-Ford (full graph, neg. edge OK):
A->A=0  A->B=2  A->C=4  A->D=7  A->E=9
Negative weight cycle detected: False
```

Floyd-Warshall All-Pairs Shortest Path Matrix and Comparison Summary:

```
3. Floyd-Warshall - All-Pairs Shortest Path (WaterGrid Hubs)

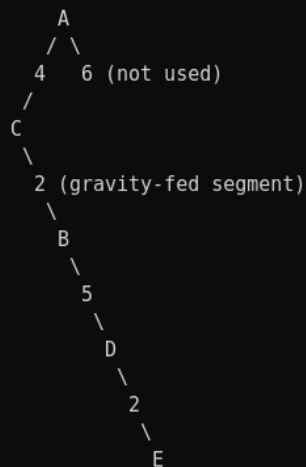
      A      B      C      D      E
A :    0      2      4      7      9
B :  INF      0     INF      5      7
C :  INF     -2      0      3      5
D :  INF     INF     INF      0      2
E :  INF     INF     INF     INF      0

Comparison Summary (A -> D shortest pumping cost):
Dijkstra (pos. subgraph) : 7
Bellman-Ford (full graph) : 7
Floyd-Warshall (full graph): 7

Bellman-Ford & Floyd-Warshall correctly use the
gravity-fed pipeline C->B (A->C->B->D = 4-2+5 = 7).
Dijkstra, run on the trimmed graph, cannot include this
segment and is unsuitable whenever negative-cost
pipelines exist in the WaterGrid network.
```

Resulting Shortest Path Tree from Source Hub A:

WaterGrid Shortest Path Tree from Source Hub A:



Path A->C->B->D->E = 4 + 2 + 5 + 2 = 9
Minimum Pipeline Cost (A to E) = 9 units

In this case study, we extended WaterGrid, the Smart Water Distribution & Monitoring System, by implementing Dijkstra's, Bellman-Ford, and Floyd-Warshall shortest-path algorithms on the same hub-and-pipeline network used for MST analysis. Water distribution hubs are represented as graph vertices, and pipelines are represented as directed weighted edges, including a negative-weight gravity-fed segment. Dijkstra's Algorithm efficiently computed shortest paths on the non-negative subgraph but was shown to be unsuitable once negative-cost pipeline segments were introduced. The Bellman-Ford Algorithm correctly handled the negative-weight edge and confirmed the absence of a negative-weight cycle in the network, while the Floyd-Warshall Algorithm computed all-pairs shortest distances in a single tabulated pass, agreeing exactly with Bellman-Ford's results. Together, the three algorithms demonstrate complementary strengths within WaterGrid: Dijkstra for fast single-source flow-path computation on standard pumped pipelines, Bellman-Ford for single-source routing where gravity-fed or negative-cost segments may exist, and Floyd-Warshall for complete all-pairs route planning across the entire water distribution network.

GitHub Link: <https://github.com/ritwikxdevv/DSA-CASE-STUDIES>

Student Signature	Faculty Signature
-------------------	-------------------